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Balancing factoid retrieval and candidate recommendation to identify toponym origins in gazetteers

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ABSTRACT

This paper investigates the adaptation of large language models (LLMs) for gazetteer enrichment, specifically to resolve missing place-name origins in the Melbourne gazetteer (Australia). Gazetteers are foundational databases for geographic information systems. However, querying toponym etymologies remains challenging: populating such attributes resists rule-based inference and necessitates direct semantic retrieval from unstructured resources. We adopted two search strategies: *factoid retrieval* for explicit origins, and *candidate recommendation* otherwise. The use of nonspatial models risks degraded spatial matching performance. Hence, we developed a retrieval-augmented generation (RAG) architecture within a geographic question-answering framework. Our system employed parallel semantic-spatial rankers that deliberately enforced a geographic bias toward target regions. Our system improved spatial ranking scores by up to 7% in normalized discounted cumulative gain. However, at the final stage, LLMs still struggled to reliably filter spatial matches. To investigate this limitation, we conducted a probing experiment on standalone LLMs. We revealed intrinsic spatial distortions regarding adjacency relations in Greater Melbourne, notably an inherent popularity bias that prevented fine-grained responses. Our findings suggest that effectively deploying LLMs for spatial data origins must advance beyond *factoid retrieval* by also refining *candidate recommendation* at every stage: calibrated biases adapt language model search to local specificities.

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Geographic information retrieval; toponym origin; spatial bias; place names; qualitative spatial relations

1. Introduction

Geographic names—defined as a Global Fundamental Geospatial Data Theme¹ by the United Nations—are *foundational* data: any derived property is a spatial axiom. Hence, knowledge discovery related to a name is likely to lead to failure if inferred with

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reasoning or predictive models. For example, in Melbourne, the records of the Victorian Gazetteer show naming idiosyncrasies behind many place names (toponyms), often represented by a single word. *Queen Street* commemorates *Queen Adelaide of Saxe-Meiningen*. A few hundred meters away, *King Street* is in fact not named after her husband, King William IV, but after *Governor Philip Gidley King*, the third Governor of New South Wales.

Missing origins (also referred to as toponym etymologies) in gazetteers are common and require linking to other documents and data sources to find the first mentions of a place name in the literature. To this end, language models are increasingly used as human-computer interfaces to receive a textual prompt involving space and match with textual resources. The recent evolution of large language models (LLMs) extends conventional spatial query techniques with new capabilities: language models shift the paradigm of structured queries, commonly operated in GIS, to information-retrieval-based question-answering (IR-QA), by bypassing explicit reasoning with text similarity. Answers to geographic queries may yield varying levels of confidence: robust retrieval necessitates two major search strategies. *Factoid retrieval* (Li *et al.* 2021, Jurafsky and Martin 2025) aims to find information stored in a database that precisely matches a user's question. When exact matches are not possible, *candidate recommendation* aims to identify and rank plausible—though not necessarily entirely correct—answers. Beyond geographic names, the use of language models offers broader applications in GIS for spatial database construction, e.g., sentiment analysis of places following natural disasters and generation (Ilyas and Sharifi 2025), or database completion, notably for urban indicators (Pan *et al.* 2025).

However, the straightforward use of language models in GIS applications masks a fundamental concern: how do spatially agnostic models (e.g. language models) filter spatial information throughout sequential pipelines?

The key contributions of our paper are summarized as follows:

1. **Enhanced spatial matches for gazetteer enrichment via *factoid retrieval* and *candidate recommendation*.** We proposed a retrieval-augmented generation (RAG) architecture and prioritized target regions via multidimensional ranking. We validated this architecture on the specific task of identifying street name origins in Melbourne and demonstrated that spatial discrepancies characterized retrieval based on language models.
2. **Evaluation of spatial distortions in LLMs.** We empirically probed adjacency knowledge across multiple LLMs, documenting asymmetries, popularity biases, and spurious spatial connections that imposed performance ceilings on spatial retrieval at fine-grained levels.

In the remainder of this paper, [Section 2](#) reviews related work in toponym enrichment and the use of LLMs in knowledge discovery from unstructured texts. [Section 3](#) defines our RAG framework that injected a spatial bias to focus on a specific area of interest. [Section 4](#) reports the results on placename origin determination, with a distinction between *factoid retrieval* and *candidate recommendation* strategies. [Section 5](#) probes standalone LLMs to explain performance ceilings, notably due to inherent

spatial biases. Finally, [Section 6](#) concludes the paper with a discussion of the implications and future directions.

2. Related work

In [Section 2.1](#), we provide an overview of heterogeneous approaches in gazetteer enrichment, based on retrieving information or querying textual resources. In [Section 2.2](#), we explore how language models can be used to identify spatial matches in textual resources.

2.1. Approaches to gazetteer enrichment

Gazetteers are fundamental spatial databases. Although gazetteers are authoritative, the authenticity of place names might be disputed and lead to place renaming (Williamson [2023](#)), notably in post-colonial regions. Determining their origin is a key step to enhancing trust in the quality of the data (Keßler *et al.* [2009](#)) with two directions: either determining a custodian, or an etymology—which is the intent of the paper. For gazetteer enrichment, intensive research has been conducted in extracting historical toponyms (DeLozier *et al.* [2016](#)), or defining the semantics of place (Kuhn *et al.* [2021](#)). In parallel, advances in open-domain information extraction (Etzioni *et al.* [2008](#)) and question-answering (Zhang *et al.* [2023](#)) foster knowledge discovery, and can be used to extract cues related to the original provenance of a gazetteer assertion.

The capacity to find an answer with a relevant rationale defines the “answerability” of a query (Rajpurkar *et al.* [2018](#), Reyes-Montesinos *et al.* [2025](#)). In this paper, we define *factoid retrieval* (Jurafsky and Martin [2025](#)) as a strategy to extract existing, correct answers from a source. However, in cases where ground truth is ambiguous, *candidate recommendation* serves as an alternative strategy to identify and score the best potential matches.

Geographic queries possess an inherent complexity (Mishra and Jain [2016](#), Mai *et al.* [2021](#)) that affects answerability—notably due to language variability. Hence, a reliable response requires predefining computable priors (as preexisting conditions) that define a *specific context of resolution*. We define a *prior* as a preliminary assumption or knowledge that affects the search. For example, the latent knowledge of pre-trained models acts as priors to enhance the answerability of queries.

2.2. Responding to spatial queries with language models

Traditionally, GIS solves geographic queries by parsing logic. Defining a logic enhances interpretability but faces one limitation: there is no guarantee that the logic exhaustively captures all decision factors, potentially leading to biased conclusions. For example, Sherman *et al.* ([2025](#)) noted that while fine-tuning helps LLMs parse geographic queries into GIS code, LLMs still fail to capture the full diversity of GIS tasks; generalization to new logic is an open challenge.

Alternatively, *generative models* rely on tunable models to memorize information, where prompts act as an instruction-based prior to highlight the most probable

answers. Here, the distinction between retrieval and recommendation is more confusing with language models, notably in question-based recommendation systems (Gupta *et al.* 2020, Zou *et al.* 2020). The generative LLMs have acquired a prior representation of the physical world (Roberts *et al.* 2023), albeit with distortions. Recent studies highlighted these limitations: Renshaw *et al.* (2025) exposed spatial biases in neighbor concepts across the USA in GPT-4 and Gemini Pro; (Fulman *et al.* 2024) identified human-like biases in cardinal relationships; Hochmair *et al.* (2024) observed limited capabilities when spatial ability is involved.

More generally, these generative methods fall into *information-retrieval-based question-answering* (IR-QA). IR-QA encodes heterogeneous data into a unified vector space, performing similarity matching via algebraic operations. Retrieval-augmented generation (RAG) (Lewis *et al.* 2021) is a computationally optimized architecture that combines a multivector retrieval system (using text encoders) with an LLM as the final analyzer. However, applying RAG to GIS presents challenges. First, LLMs do not thoroughly capture all spatial regularities (Janowicz *et al.* 2025). Second, the knowledge within pretrained foundation models is static and difficult to update. In an ever-changing world, predictive models rapidly become obsolete, and machine unlearning—or editing—becomes increasingly challenging given the billion-parameter scale of modern LLMs (Li *et al.* 2024).

The integration of neural language models represents a fundamental shift in GIS applications. The internal knowledge of these models acts as a statistical prior, effectively substituting explicit logic with the probabilistic outputs. However, this flexibility comes at the cost of spatial distortions due to predictive biases (Swinton 1981).

3. Enhancing spatial search in texts via parallel ranking

This section introduces a unified architecture using retrieval-augmented generation (RAG) in Figure 1. We simultaneously conducted *factoid retrieval* and *candidate recommendation*. This model combined two objectives—semantic and geographic matches—and the architecture was deliberately biased toward designated target regions to qualitatively enhance spatial anchoring. In this regard, our method aligned conceptually with the multiview representation learning in deep learning (Li *et al.* 2019), where complementary perspectives are jointly modeled to enhance the system performance.

3.1. Problem formulation

Given a question X that involves a spatial concept, geographic information retrieval aims to identify the correct answer y from a set of documents, given as input. We formulated the geographic information retrieval system as a question-answering problem, where both the input question X and the retrieved answer y are in natural language.

We illustrate the problem with the retrieval of placename origins: the question $\{X: \text{"Who is Batman Avenue in Melbourne named after?"}\}$ focuses on the location "Batman Avenue, Melbourne" with the keyword $\{X_{\text{root}}: \text{"Batman"}\}$ and expects the

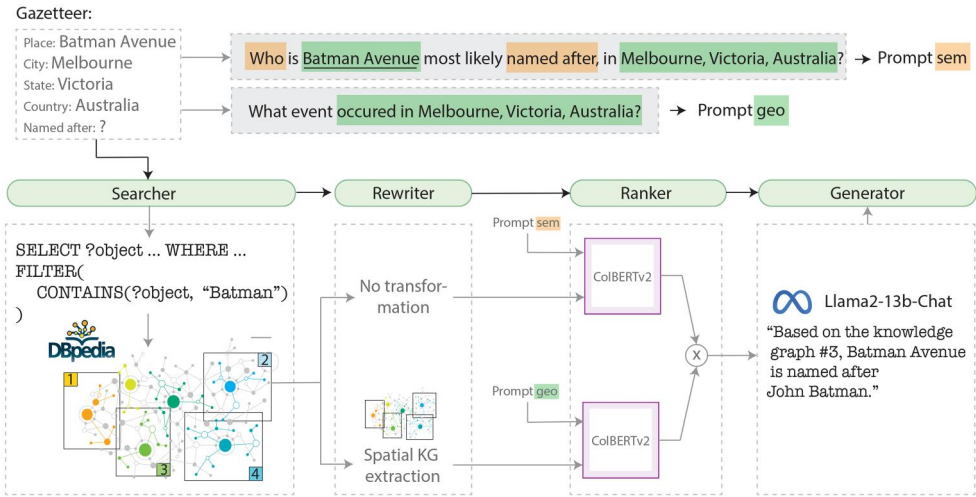


Figure 1. Description of the model V2 based on a double subranker and two views after the rewriter. With only one view of the input data, the model V1 has the same architecture as V2 but without the rewriter module.

answer $\{y: \text{"According to Wikipedia, Batman Avenue is named after John Batman"}\}$. Finally, the concise answer to the initial question is $\{y_{\text{pred}}: \text{"John Batman"}\}$.

To tackle this problem, a critical challenge is to comprehend and process spatial information that is embedded in an unstructured textual format. We leveraged an external spatial knowledge graph to address this challenge. Knowledge graphs store spatial facts in a structured format: a set of directed triplets in the form of (s, p, o) , where s represents the subject entity, o represents the object entity, and p represents the predicate (i.e., relation). Each triplet represents a factual statement asserting that the subject s is related to the object o via the relation p . An example triplet is: $(s: \text{John_Batman}, p: \text{commemorated_by}, o: \text{Batman_Avenue})$. With a knowledge graph, the process of answering a specific spatial question employed a RAG framework to select relevant nodes and generated a refined answer.

We revisit existing architectures for spatial RAG systems in Section 3.2, and introduce our model in Section 3.3.

3.2. Architectures in geographic information retrieval

To build GeolR systems that support natural language interaction, modern architectures employ modular designs, notably with the retrieval-augmented generation (RAG) framework. This typically consists of an *indexer* to efficiently store all available documents, a *ranker* to retrieve relevant documents, and a *generator* that synthesizes an answer aligned with the original query. When designing such architectures, one must find a balance between the accuracy of retrieval and the efficiency (e.g. time complexity) of the algorithm. Various configurations have been proposed to meet diverse real-world scenarios, accommodating specific constraints on time budgets and hardware resources.

As a core component of RAG, the *ranker* extracts the top- k documents from a database that are most relevant to a given query. These documents form a retrieved set forwarded to the generator. In the literature, the relevance of a document with respect to a query is usually computed as the cosine similarity between their embeddings. However, rankers such as ColBERT (Khattab and Zaharia 2020) are typically trained on general knowledge corpora and possess limited capability to analyze spatial knowledge.

LLMs are used as *generators* for final GeolR responses. Indeed, without further training, generative LLMs have demonstrated remarkable suitability for this task (e.g., Petroni et al. 2019, Cohen et al. 2023, Wei et al. 2023).

3.3. Our proposed model

Inspired by the RAG architecture, our proposed model was designed as a modular framework consisting of four main components: (1) a *searcher* module, which converted the question X into a structured query over DBpedia and extracted a sub-knowledge graph with candidate triplets; (2) a *rewriter* module, which transformed the lengthy and unstructured descriptions from the retrieved triplets into a normalized and structured format; (3) a *ranker* module, which ranked the triplets based on their semantic and spatial relevance to the question; and (4) a *generator* module, which synthesized a natural language answer from all the previously retrieved information. Figure 1 presents an overview of the overall architecture with the model **V2**: each module filtered a decreasing number of candidates (top- k_{searcher} , top- k_{ranker} and top- $k_{\text{generator}}$ respectively).

Searcher module

For a question X^i , the *searcher* extracted potential candidates (indexed as j) from the knowledge graph DBpedia. Each candidate was described by one sub-knowledge-graph (referred to as document $D_{i,j}$) where all the triplets shared the same subject s .

Rewriter module

When analyzing the extracted knowledge graph, certain extracted triplets possessed long textual descriptions with insufficiently parsed geographical knowledge. Through the *rewriter* module, we derived the document $D_{i,j}$ into another document-knowledge graph $D_{i,j}^{\text{geo}}$ by extracting all the spatial triplets from $D_{i,j}$. This step was performed first with open information extraction over the document-knowledge graph $D_{i,j}$, and second with an RML mapping (Dimou et al. 2014) that exported the document $D_{i,j}^{\text{geo}}$ as an RDF/XML knowledge graph.

Ranker module

To rank the documents $D_{i,j}$ and $D_{i,j}^{\text{geo}}$ according to the question X^i , we decomposed the *ranker* module into two subrankers that ran in parallel for semantic and geographical ranking; both subrankers shared the same parameters. As previously discussed, pretrained language models struggle to capture geographical knowledge. As a solution, we used the fine-tuned version of ColBERTv2 (Santhanam et al. 2022) developed

by Horde Vo *et al.* (2026); we refer to their work for more details. This version—denoted as the ranker of **C2a**—was trained on (a) 65,048 question-answer pairs over DBpedia and derived from the QALD9 dataset (Usbeck *et al.* 2018) and (b) on 26,401 question-answer pairs related to spatial adjacencies in GeoNames across the world. We performed a late fusion of the two subranking scores to generate the final ranking of all candidates, as a simulated probability over two cumulative conditions:

1. the condition of semantic alignment (*sem*): the document $D_{i,j}$ contains information that—partially or totally—answers the question X_i ;
2. the condition of geographic alignment (*geo*): one of the locations mentioned in the document $D_{i,j}$ is spatially contained in the location in the question X^i . For example, a document mentions “Melbourne”; the question mentions “Australia.”

The probability is calculated as:

$$P(\text{sem} \cap \text{geo}, X^i, D_{i,j}) = P(\text{sem}, X^i, D_{i,j} | \text{geo}) \times P(\text{geo}, X^i, D_{i,j}) \quad (1)$$

To emphasize the importance of space in fusion, we used the threshold hyperparameter ($\rho \in [0, 1]$) to filter out the candidates that were not geographically related:

$$P(\text{sem} \cap \text{geo}, X^i, D_{i,j}) \simeq \begin{cases} P(\text{sem}, X^i, D_{i,j}) \times P(\text{geo}, X^i, D_{i,j}^{\text{geo}}) & \text{if } P(\text{geo}, X^i, D_{i,j}^{\text{geo}}) \geq \rho \\ 0 & \text{if } P(\text{geo}, X^i, D_{i,j}^{\text{geo}}) < \rho \end{cases} \quad (2)$$

In practice, the probabilities were calculated as a similarity score between two texts. The first text was (a) a prompt that wrapped a given question X^i with the condition (*sem* or *geo*) via a template. The second text was (b) the body of a document ($D_{i,j}$ or $D_{i,j}^{\text{geo}}$). The templates are presented in Annex B.

We filtered the top- k_{ranker} most relevant documents according to the fused ranking scores.

Generator module

As a final step, the *generator* module received the question X_i , then interpreted the selected set of top- k_{ranker} documents $D_{i,j}$ and produced the final answer.

4. Experiment 1: application on gazetteer enrichment

We evaluated our model described in Section 3 on one downstream task: gazetteer enrichment to identify the origin of street names in Melbourne (Australia). We analyzed critical decision points within the processing pipeline to ensure spatially trustworthy responses, via a *factoid retrieval* or *candidate recommendation* strategy.

4.1. Materials

As a gold reference, we used the gazetted register *Vicnames*² (Gazetteer from Geographic Names Victoria, Australia), consisting of 248 street names in Melbourne City with an authoritative annotated origin.

In terms of discrimination, the pipeline selected 10K, 10 and 1 items after the searcher, the ranker and the generator, respectively. In the late fusion of the ranker, we set $\rho = 0$ as this value produced the best empirical results.

Across three main components, we selected: an AWQ-quantized Gemma-2-9B (Team *et al.* 2024, Lin *et al.* 2024) for query rewriting, a finetuned ColBERTv2 architecture (Santhanam *et al.* 2022) for candidate scoring, and an AWQ-quantized Llama-2-13B (Touvron *et al.* 2023) for response generation. We served both LLMs via the vLLM framework (Kwon *et al.* 2023) to accelerate inference. The prompts are presented in Annex B.

We proposed two models for comparison. First, the model **C2a** defined in Section 3.3 was used as a baseline after the ranker and the generator. Second, we also used the commercial model **GPT-4o-mini** without an access to external databases. The model GPT-4o-mini is a standalone LLM (generator only): consequently, the model GPT-4o-mini was used as a baseline after the generator only.

4.2. Evaluation protocol

Evaluation perspectives. System performance was evaluated at two stages: after the ranker and after the generator. Language models interpret any textual condition as a soft constraint, either based on a semantic alignment (*sem*), or on a geographic alignment (*geo*) (Section 3.3). To study spatial hierarchy in candidate recommendation, we differentiated the geographic condition *geo* into two subconditions: *geo_aus* and *geo_vic*. In summary:

- *sem*: the retrieved document contains information that—partially or totally—answers the question;
- *geo_aus*: the retrieved document mentions a location related to Australia;
- *geo_vic*: the retrieved document mentions a location related to Victoria (Australian state containing Melbourne).

Metrics. For each of the conditions *sem*, *geo_aus*, *geo_vic*, we calculated the following metrics:

- the *mean reciprocal rank* (MRR): a score close to 1.0 indicates that the first relevant candidate appears close to rank 1;
- the averaged *normalized discounted cumulative gain* (nDCG): a score close to 1.0 indicates that relevant candidates are ranked higher than any irrelevant candidate;
- the *precision* (P), which indicates how many relevant candidates on average are recommended in the top-*k* set;
- the *hit ratio* (HR), which indicates the frequency with which a set of results contains at least one relevant recommendation.

We refer to Annex C for the mathematical definitions used to compute the metrics.

Dataset partition. For 93 streets (38%) in the dataset DS:Gazetteer, the searcher successfully extracted a knowledge graph $\mathcal{KG}_i$ containing at least one triplet related to

the gazetted origin ($n = 93$ samples). This subset is referred to as *restricted DS:Gazetteer*, where the model was expected to adopt a *factoid retrieval* strategy. The metrics restricted to this subset are denoted with *, e.g., MRR^* . For the other entries in DS:Gazetteer, the model followed a *candidate recommendation* strategy.

Global performance of the model. As the generator was the final module of our GIR model, the *global precision of the model* was equal to the hit ratio after the generator with $n = 248$ samples. Analogously, the *global recall of the model* was equal to the hit ratio after the generator (HR^*) restricted to the streets whose origin was mentioned in the knowledge graph after the searcher.

To study the impact of the rewriter module, we also implemented two variants of our proposed model: one denoted as **V1** without the rewriter module, and another denoted as **V2** with the rewriter module.

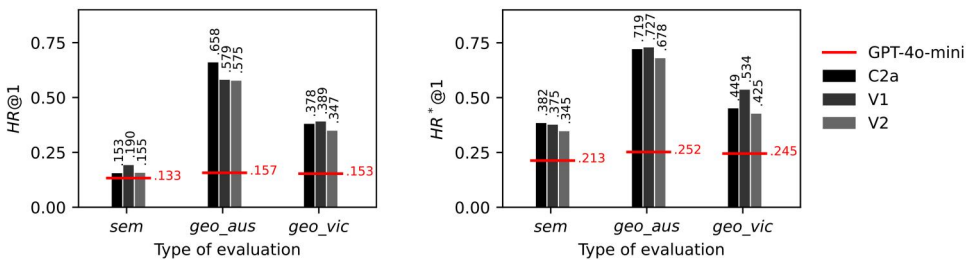
Annotators. The evaluation after the ranker was conducted automatically, and after the generator by one annotator (the lead author) in comparison with the annotated Victorian Gazetteer.

4.3. Results

Our results showed that the overall determination of an origin reached a performance ceiling, at the frontier between automated information extraction and humanities (Section 4.3.1). However, our computational methodology demonstrated significant improvements in reducing the loss of spatial data throughout a sequential pipeline (Section 4.3.2).

4.3.1. Performance ceiling in the determination of place name origins

Figure 2 compares the final results after the generator with the hit ratio HR (also the precision of our GIR model) given for the models V1, V2. The models C2a, GPT-4o-mini were used as baselines. The hit ratio HR_{sem} (also the recall of our GIR model) was similar across the models C2a, V1, V2 and GPT-4o-mini (Figure 2(a)). The restricted hit ratio



(a) Hit ratio (HR) after the generator with the whole dataset ($n = 248$); HR is also the global precision of our GIR model.

(b) Hit ratio (HR^*) restricted to the streets where the knowledge graph mentions an origin ($n^* = 93$) after the generator; HR^* is also the global recall of our GIR model.

Figure 2. Evaluation of the hit ratio (HR) on the matchings over the conditions *sem*, *geo_aus* and *geo_vic* after the generator for the GIR models C2a, V1, V2. The commercial GPT-4o-mini—used as a standalone LLM—was also used for benchmark. When evaluated on the whole dataset (resp. restricted dataset), the hit ratio is equal to the *global precision of the model* (resp. *global recall of the model*).

HR_{sem}^* showed higher performance for the models C2a, V1 and V2 (Figure 2(b)) than GPT-4o-mini.

The nature of errors differed across the models. Contrary to the models C2a, V1 and V2, the baseline model GPT-4o-mini provided more correct answers for place names derived from common objects that were missing in the extracted knowledge graphs, e.g. such as Plover Lane (a bird), Bridleway Walk (a type of way), or Ti Tree Walk (a plant). In contrast, our proposed models provided more specialized outputs than GPT-4o-mini but reduced the autonomy of language models to generalize.

In comparison with the model C2a, the results for the models V1 and V2 after the generator were inconsistent: no clear pattern emerged in Figure 2.

The observed performance ceiling stemmed from the absence of explicit provenance mentions in the source data, compounded by the inability of computational approaches using LLMs to *infer* axiomatic attributes such as an origin. However, the final decision-making process remained highly dependent on the quality of the information filtered by the ranker, particularly if human validators replaced the LLMs.

4.3.2. Limiting spatial information losses

Retrieving an origin was the primary objective of the search, whereas finding spatial matches served as a complementary objective. Hence, one major architectural optimization was to balance the importance of both search objectives.

The ranker was the main retrieval module, selecting 10 candidates from 10k triplets; the results in this section are presented in two categories. First, Table 1 reports the averaged mean reciprocal rank; the averaged normalized discounted cumulative gain; the hit ratio; and the mean average precision for the models C2a, V1, and V2 after the ranker module. On the restricted DS:Gazetteer dataset, we expected the model to follow a *factoid retrieval* strategy: the model should not discard any relevant information, with a focus on the conditions related to *sem*. Second, Table 2 presents the same metrics across the whole dataset DS:Gazetteer ($n = 248$). As 63% of placenames did not have an origin in the dataset, the model should employ a *candidate recommendation*

Table 1. Evaluation of the performances on a subset of the dataset—the extracted knowledge graph (KG) from the searcher must contain a mention of the origin—DS:Gazetteer ($n = 93$) of the models C2a, V1 and V2 after the ranker for the different types of observations (relevance of the semantic evaluation *sem*, or spatial evaluations *geo_aus* or *geo_vic*).

Ranker		... restricted to the streets where the KG mentions an origin ($n^* = 93$)			
$k_{ranker} = 10$					
Model	Condition	MRR*	nDCG*	P*	HR*
C2a	sem	.431	.382	.173	.506
	geo_aus	.889	.909	.664	.944
	geo_vic	.740	.745	.366	.798
V1	sem	.475 (+10%)	.436 (+14%)	.177 (+2%)	.523 (+3%)
	geo_aus	.888 (+0%)	.930 (+2%)	.707 (+6%)	.955 (+1%)
	geo_vic	.723 (−2%)	.754 (+1%)	.419 (+14%)	.784 (−2%)
V2	sem	.467 (+8%)	.428 (+12%)	.195 (+13%)	.506 (+0%)
	geo_aus	.891 (+0%)	.950 (+5%)	.700 (+5%)	.954 (+1%)
	geo_vic	.697 (−6%)	.737 (−1%)	.380 (+4%)	.782 (−2%)

These scores are marked with *. Each street in the dataset DS:Gazetteer represents one independent experiment; results were averaged on 93 elements. The relative differences were calculated relative to the model C2a.

Table 2. Evaluation of the performanc on the dataset DS:Gazetteer ($n = 248$) for the models C2a, V1 and V2 after the ranker for the different types of observations (relevance of the semantic evaluation *sem*, or spatial evaluations *geo_aus* or *geo_vic*).

Ranker		Scores on DS:Gazetteer ($n = 248$)			
$k_{\text{ranker}} = 10$					
Model	Condition	MRR	nDCG	P	HR
C2a	sem	.229	.162	.076	.221
	geo_aus	.780	.858	.585	.892
	geo_vic	.602	.657	.306	.748
V1	sem	.253 (+10%)	.191 (+18%)	.084 (+11%)	.244 (+1%)
	geo_aus	.797 (+2%)	.883 (+3%)	.635 (+9%)	.900 (+1%)
	geo_vic	.637 (+6%)	.693 (+5%)	.369 (+21%)	.747 (+0%)
V2	sem	.250 (+9%)	.184 (+14%)	.091 (+20%)	.228 (+3%)
	geo_aus	.800 (+3%)	.903 (+5%)	.637 (+9%)	.904 (+1%)
	geo_vic	.643 (+7%)	.703 (+7%)	.347 (+13%)	.749 (+0%)

The relative differences were calculated relative to the model C2a.

strategy to exhibit the most relevant candidates with the highest scores in terms of spatial matching (i.e., conditions related to *geo_aus* and *geo_vic*).

In comparison with the model C2a, the separate treatment of spatial understanding with two rankers increased most of the spatial scores in [Tables 1](#) and [2](#). As a qualitative visualization, [Figure 3](#) displays the first 15 places in the dataset DS:Gazetteer and visualizes which results from the top-10 were linked to the country of Australia or the state of Victoria. We observed that the models V1 and V2 sorted spatial information with better qualitative results and a better geographical discrimination. Candidates with higher ranks (close to 1) were more frequently related to Australia or Victoria than with the model C2a. Indeed, the precision $P_{\text{geo_aus}}$ increased by +9% with the model V2 on the dataset DS:Gazetteer. In addition, the sorting was more homogeneous: the model V2 improved $n\text{DCG}_{\text{geo_aus}}$ by +5%.

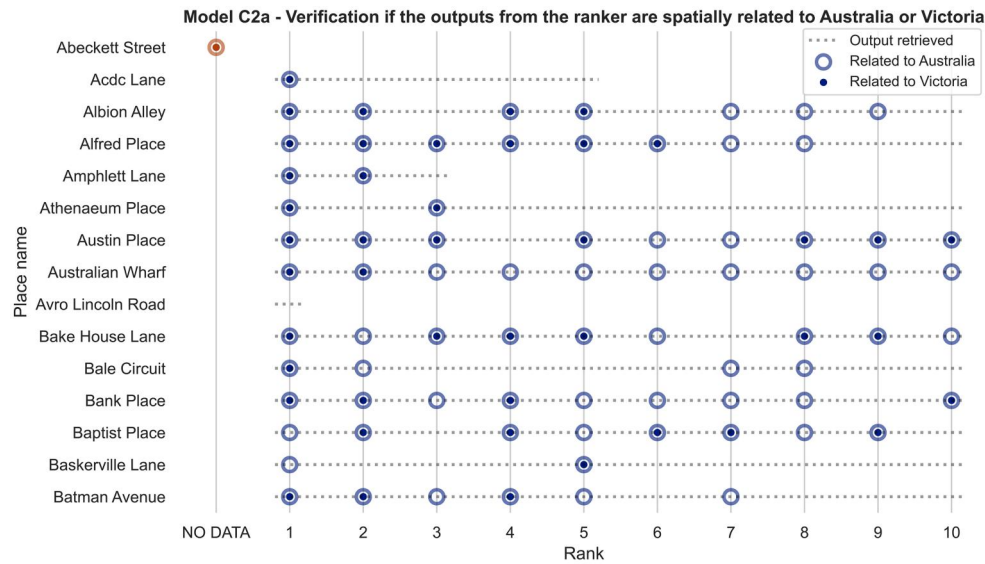
On the restricted subset of the dataset DS:Gazetteer, spatial scores (*geo_aus* and *geo_vic*) lacked a clear trend. In contrast, the semantic scores (*sem*) were higher across all metrics for the models V1 and V2 than for the model C2a. Indeed, the models focused on *existing* origins, which could actually be uncorrelated with the target areas.

The rewriter module was responsible for extracting a spatial sub-knowledge-graph to avoid information overload and focus on the spatial objective. As shown in [Table 2](#), the model V2 demonstrated a marginal improvement in nDCG across all three conditions on the DS:Gazetteer dataset compared to the model V1.

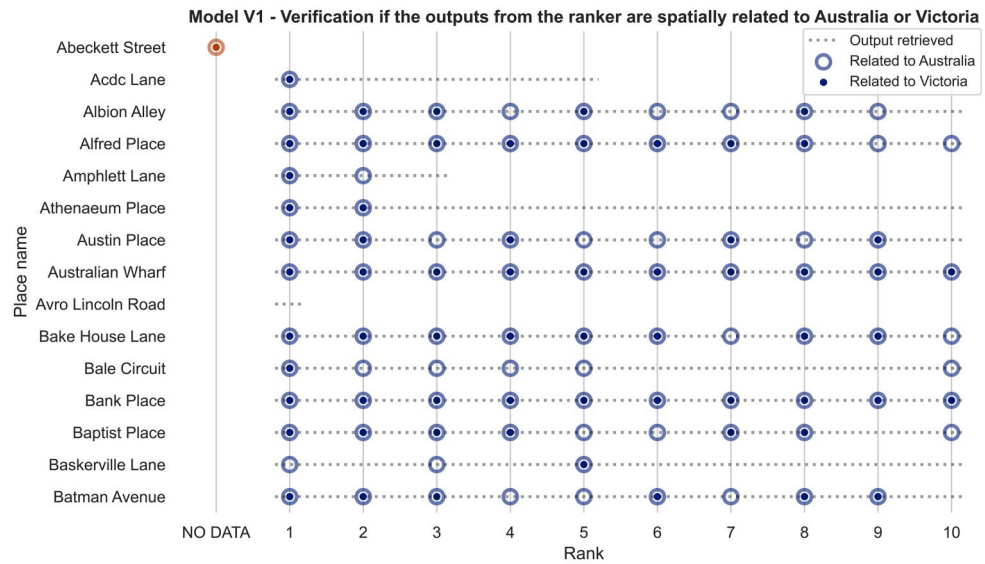
However, the model V2 did not significantly outperform the model V1 in our specific architecture. If we worked on short contexts (a chunk of 512 tokens), then the information contained in the input knowledge graph $D_{i,j}$ and its derived knowledge graph $D_{i,j}^{\text{geo}}$ was equally rich. In addition, both knowledge graphs $D_{i,j}$ and $D_{i,j}^{\text{geo}}$ yielded similar retrieval scores.

4.4. Discussion

Through our ablation study, we observed three key lessons in using language-model-based architectures to identify the origin of a placename.



(a) Model C2a with no rewriter and on ranker



(b) Model V1 with no rewriter and two subrankers

Figure 3. Retrieved items for 15 place names after the ranker module regarding *geo_aus* and *geo_vic*. Fine-tuning is expected to highlight spatially related candidates at the highest ranks. This property is characterized by the normalized discounted cumulative gain (nDCG).

4.4.1. Parallel ranking breaks down complexity in spatial queries

Finetuning and rewriting are common NLP techniques to reduce ambiguity and enhance the domain specificity of models. However, these techniques proved insufficient for geographic questions. Instead, the inherent complexity of geographic questions requires more guidance in managing the current access to existing resources.

As previously discussed (Section 2), a single language model inherently possesses strong priors. To recalibrate their influence, employing parallel ranking was the primary mechanism to *rebias* the model toward stronger geographic grounding, with soft constraints. The scope of the case study was constrained to Melbourne, as the main focus was to explore the capabilities of language models on the highly resource-intensive task of open-domain information extraction. In Appendix D, we present an alternative experiment using parallel ranking to recommend points of interest with a controlled input dataset, generalizing our observations.

Nonetheless, certain biases of language models still remained strongly anchored. For example, the model might find a—correct—link between ACDC Lane and the famous rock band AC/DC from Sydney (New South Wales), or *Niagara* Lane in Melbourne to the well-known *Niagara* Falls in North America, instead of recommending a more locally plausible origin from Victoria (Australia).

4.4.2. Generalization faces spatial nonstationarity at fine levels

A posteriori, there is no guarantee that our solution is universally superior to any other, for example, GPT-4o-mini. The main reason is that there is no statistical pattern behind the distribution of placename origins. Hence, the assumption of independent and identically distributed (i.i.d.) data does not hold, which inherently challenges the generalization capabilities of any model. Applied to spatial queries, spatial nonstationarity (Brunsdon *et al.* 1996) limits the transferability of reasoning to identify placename origins: paradoxically, LLM-generated persuasive explanations could even appear more plausible than the gold rational annotations from gazetteers.

The lack of a unique, well-defined objective is common in natural language processing, characterized by a delicate interpretation of any significance metric (Søgaard *et al.* 2014). By extension, geographic queries face the same issue with undefined solutions.

However, empirical evidence suggested that our solution was better optimized *within our specific context of resolution*: finding more candidates from Victoria (Australia) rather than elsewhere. As suggested by Søgaard *et al.* (2014), an evaluation through different metrics showed a better correspondence between the outputs and *our* context. Hence, we effectively performed geographic grounding through parallel ranking for both *factoid retrieval* and *candidate recommendation*.

4.4.3. LLMs are not optimal spatial decision models but recommendation models

Under the closed-world assumption (Reiter 1978) in *factoid retrieval*, geographic queries might not find any clue in spatial databases; hence, the answer would be “*There is no origin for the place name.*” Under the open-world assumption, the origin is “*unknown.*” In both cases, traditional retrieval is ineffective for discovering new associations.

Alternatively, *candidate recommendation* allows for probable answers. In this context, the advantage of neural methods is to directly compute correlations between a placename and a candidate to bypass intermediate reasoning steps: this operation is

possible because language models rely on strong priors learned through their extensive training. In spite of coherent matches, language models also discard relevant information with a lack of transparency; hence, ranking metrics are hard to optimize simultaneously across all conditions in geographic questions.

Overall, the limited performance of LLMs at the end of the pipeline—with the generator—showed that we also reached a pragmatic limit: how far can language models perform decision-making for socially constructed topics such as place name origins? A thorough analysis of how and why an LLM produces a rationale would require a deeper study in terms of explainability. However, this objective was not the main intent of the paper.

5. Experiment 2: probing LLMs in retrieving spatial adjacencies

Section 4 showed that the final LLMs unsurprisingly failed at making decisions for societal queries. Nevertheless, there was one strength: we could automate retrieval to increase the ratio of spatially relevant candidates for toponym origins before delegating the final decision to human judgment. However, the final LLM was not exempt from introducing spatial discrepancies between an answer and the target area, and the models could not provide any self-correction either. According to Tobler's First Law of Geography (Tobler 1970), we expected answers to be spatially close to the target area. In this section, we argue that *LLMs cannot provide exhaustive spatial corrections* in complex systems because their basic knowledge of adjacency is brittle.

5.1. Problem definition

We tested the binary relation of adjacency in Greater Melbourne using gazetteers as ground truth to evaluate the upper bound of LLM performance.

5.2. Materials

We extracted a set S of $n = 330$ suburbs from the Victorian Government state gazetteer³. The subset contained suburbs within a distance of roughly 30 km around the city of Melbourne (Victoria, Australia), also referred to as 'Greater Melbourne.' The suburbs included areas of high activity, such as Melbourne CBD, as well as less well known areas, such as Diggers Rest.

The experiments were conducted with the language model ColBERTv2 (Santhanam *et al.* 2022) (110 M parameters) as an encoder, and six different decoder models representative of a range of popular LLMs: Llama2 (7B, 13B) (Touvron *et al.* 2023), DeepSeek-R1-distill-Qwen (32B), Mixtral (8×7B) (Jiang *et al.* 2024), Llama 3.3 (70B) (DeepSeek-AI *et al.* 2025), and GPT-4o-mini (unknown number of parameters) (OpenAI *et al.* 2024). These models were employed without fine-tuning, in one-shot learning, and without access to external knowledge bases.

5.3. Methodology

Using the geometry of the suburbs, a directed, symmetric graph $G_S = (V, E)$ was created, where V is the set of suburb names and $(v_1, v_2) \in E$ if the geometries of suburbs v_1 and v_2 are adjacent (i.e., satisfied the 9IM relation *touches*).

For each language model, we constructed a directed graph G_{LM} that cataloged the adjacency relationships between suburbs in the set S , derived from the model's outputs. For example, the suburb *Carlton* is in actuality adjacent to *Fitzroy*, i.e., $(Carlton, Fitzroy) \in G_S$. However, the regularities of language captured in each language model do not necessarily mirror the regularities of geographic space, captured in a gazetteer.

To quantify these inconsistencies, we prompted the decoder-only models to query which suburb was immediately adjacent to a given suburb using: "List all the suburbs that are immediately adjacent to [suburb v_1], Victoria, Australia. [Further instructions]" (see Annex A for full details of the further instructions supplied).

With the encoder-only model ColBERTv2 (Santhanam *et al.*2022), we probed the model using the query: "Which suburb is immediately adjacent to {suburb v_1 }, Victoria, Australia?" for each reference suburb $v_1 \in V$. ColBERTv2 does not have the capacity to generate an answer *ex nihilo*. Instead, ColBERTv2 requires a set of existing documents to rank; each document was populated with the template "[suburb v_2] is immediately adjacent to [suburb v_1]." To construct the adjacency graph G_{LM} for ColBERTv2, we extracted the top- k ranked candidates, where k corresponded to the ground-truth degree of suburb v_1 in G_S .

5.4. Evaluation

Annotation. We evaluated the geographic regularities captured by each language model by comparing its adjacency graph G_{LM} with the ground-truth adjacency graph G_S . For each edge in a graph G_{LM} , we labeled the edge as *true positive* (TP) if the same edge was found in G_S ; or *false positive* (FP) if the edge did not exist in G_S . For each edge in G_S but not in G_{LM} , we labeled the edge as *false negative* (FN).

Precision and recall. We reported overall *precision* and *recall* at the graph level. Precision measured the accuracy of the language model's predictions, calculated as $TP / (TP + FP)$. Recall indicated the model's ability to retrieve the correct neighboring suburbs, calculated as $TP / (TP + FN)$.

False positive error breakdown. In our experiments, false positives occurred in two distinct forms: (a) *node-level errors*, where the language model created an edge between two suburbs in which at least one of those suburbs did not exist in S ; and (b) *edge-level errors*, where the two connected suburbs existed in S , but the edge connecting them did not exist in G_S . Adapting the term "out-of-vocabulary" from the LLM literature, we treated all nodes (suburbs) S as our (placename) vocabulary. In turn, we denoted the total number of edges of Type (a) as FP_{OOV} and the total number of edges of Type (b) as FP_{-OOV} . Thus, FP_{OOV} represented edges connecting out-of-vocabulary (OOV) suburbs, and FP_{-OOV} represented false positive edges connecting in-vocabulary suburbs.

Symmetry. We also wanted to assess each language model’s ability to accord with expected geographical principles, such as symmetry of adjacency relations (i.e., if v_1 is adjacent to v_2 it implies that v_2 is adjacent to v_1). To this end, we computed the ratio of symmetrical edges, defined as the proportion of edges that were bidirectional—i.e., both (v_1, v_2) and (v_2, v_1) existed, regardless of correctness—out of the total number of edges in the generated graph G_{LM} .

Centrality. Further, we evaluated any bias toward certain geographical locations in each language model by computing the *standardized indegree centrality* for each node. For a node v , the standardized indegree centrality was computed as $C(v) = \frac{\deg_{in}(v)}{|\text{Neigh}(v)|-1}$, where $\deg_{in}(v)$ was the indegree of the vertex v in G_{LM} , $|\text{Neigh}(v)|$ was the number of adjacent vertices for v in G_S . Note that the standardized indegree centrality, the ratio of FP_{OOV} and FP_{-OOV} could be greater than 1.0. In that case, the model generated *more* edges of a given type than the existing number of edges for a suburb in the G_S .

5.5. Results

Figure 4 illustrates the adjacency graphs G_{LM} created using Llama 13B and GPT-4o-mini. Llama 13B tended to overconnect suburbs that were not in actuality neighbors, as shown by the preponderance of longer, blue “fabricated” (false positive) edges. In contrast, GPT-4o-mini seemed to capture more accurately the true positive (green edges). However, GPT-4o-mini also produced a high proportion of bicolored edges, indicating high levels of asymmetry, where the language model indicated that v_2 was not adjacent to v_1 even though v_1 was correctly indicated as adjacent to v_2 .

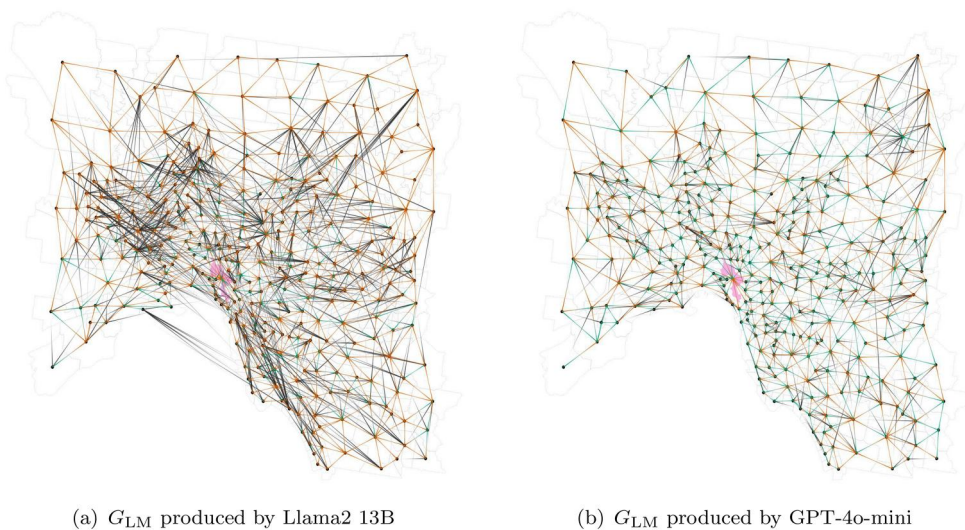


Figure 4. Sunbeam visualization of G_{LM} produced by Llama2 13B and GPT-4o-mini for the Melbourne metropolitan area. The color of an edge as it leaves a node indicates its category. True positives are indicated in (green); false negatives are (orange); and false positives in (gray). Melbourne City is in (pink).

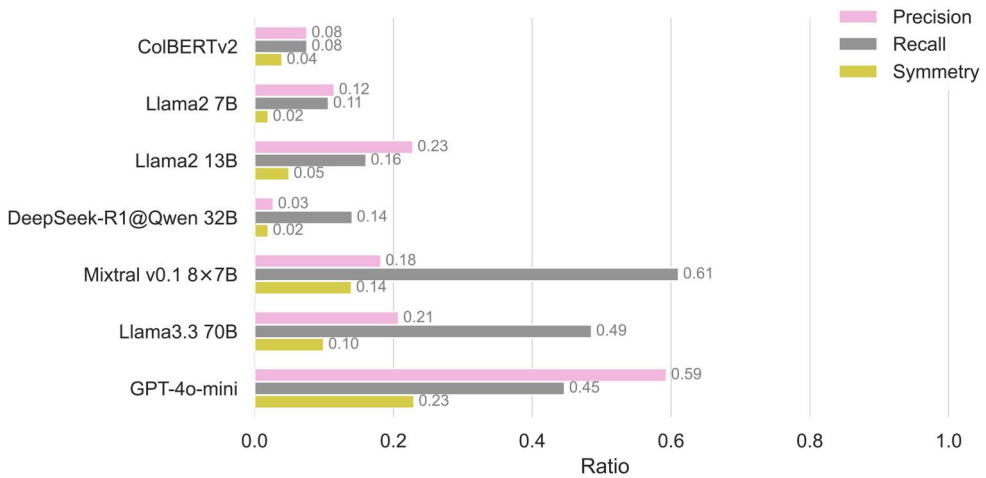


Figure 5. Precision, recall and ratio of symmetry for the graphs G_{LM} .

5.5.1. Graph-level irregularities

Figure 5 presents the overall precision and recall for the seven compared language models. In general, the precision was low across all models with 0.59 for the best performing model in terms of precision: GPT-4o-mini. In terms of recall, 4 out of the 7 models (ColBERTv2, Llama2 7B, Llama2 13B and DeepSeek-R1@Qwen 32B) exhibited performance below 0.2.

To better understand the source of errors, Figure 6 shows the ratios of false positives of Type (a) and the ratio of false positives of Type (b) for all language models. Respectively, these ratios were computed as $FP_{OOV}/|G_S|$ and $FP_{-OOV}/|G_S|$. Any ratio greater than 0.0 indicated the existence of false positive edges produced by a language model, i.e., edges fabricated by language models as so-called “hallucinations.” In general, all language models tended to fabricate many edges: DeepSeek-R1@Qwen 32B created five times as many non-existing as existing edges (split between 2.69 times in-vocabulary edges, FP_{-OOV} , plus 2.31 times out-of-vocabulary edges, FP_{OOV}). Figure 6 also showed that all language models have higher ratio of FP_{-OOV} than FP_{OOV} . This indicated that more errors originated from creating edges (FP_{-OOV} edges) between known suburbs rather than from creating non-existing nodes (involved in the FP_{OOV} edges), i.e., errors mainly occurred at edge-level.

5.5.2. Suburb-level irregularities

Figures 7 and 8 present the suburb-level breakdown of precision and recall for all language models. The distributions of precision and recall varied significantly across different suburbs. Larger language models, such as GPT-4o-mini and Llama 3.3 70B, tended to exhibit overall better performance in precision and recall. Nonetheless, the suburb-level breakdown showed that the generated outputs could be extremely inaccurate for some suburbs, reaching precision and recall below 0.25.

Most higher performing models (such as Mixtral 8×7B and Llama 3.3 70B) achieved higher recall at the cost of reduced precision. To highlight this pattern, Figure 9 shows the suburb-level distribution of the ratio of false positives to true positives. In general

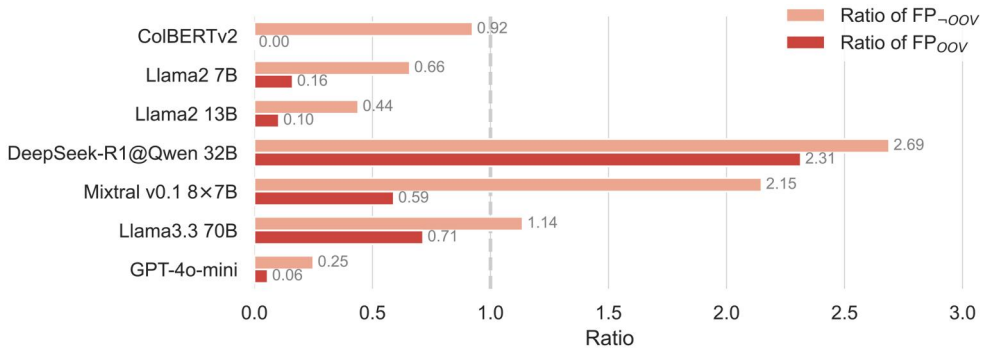


Figure 6. Ratio of false positive edges connecting “out-of-vocabulary” (OOV) suburbs (FP_{OOV}) and “in-vocabulary” suburbs (FP_{OOV}). A ratio greater than 0.0 indicates the existence of false positive edges. A ratio greater than 1.0 indicates that a language model generated more fabricated edges than the ground-truth edges in G_S .

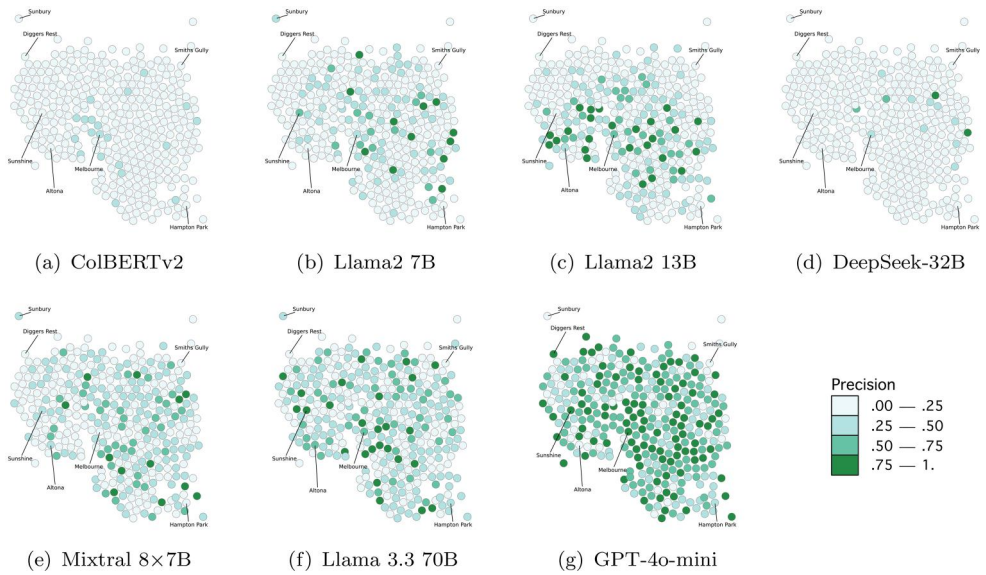


Figure 7. The suburb-level breakdown of *precision* for all compared language models.

we observed that models with increasing numbers of parameters tended towards increasing high rates of false positives. One reason for this trend was the tendency of larger models to fabricate more false positives. For example, DeepSeek-R1-Distill-Qwen 32B fabricated non-existing suburbs “Heidelberg South”, “Heidelberg North”, or “Heidelberg Village” as neighbors of the existing suburb “Heidelberg.”

GPT-4o-mini was a likely exception to this pattern, as it was assumed to have a large number of parameters but exhibited a false positive ratio below 0.25. Even with this high-performing model, however, the suburb-level distribution still exhibited uneven variances across different suburbs.

We also evaluated whether the graph G_{LM} generated by each language model respected the symmetry of adjacency relations. Figure 5 already showed that all tested

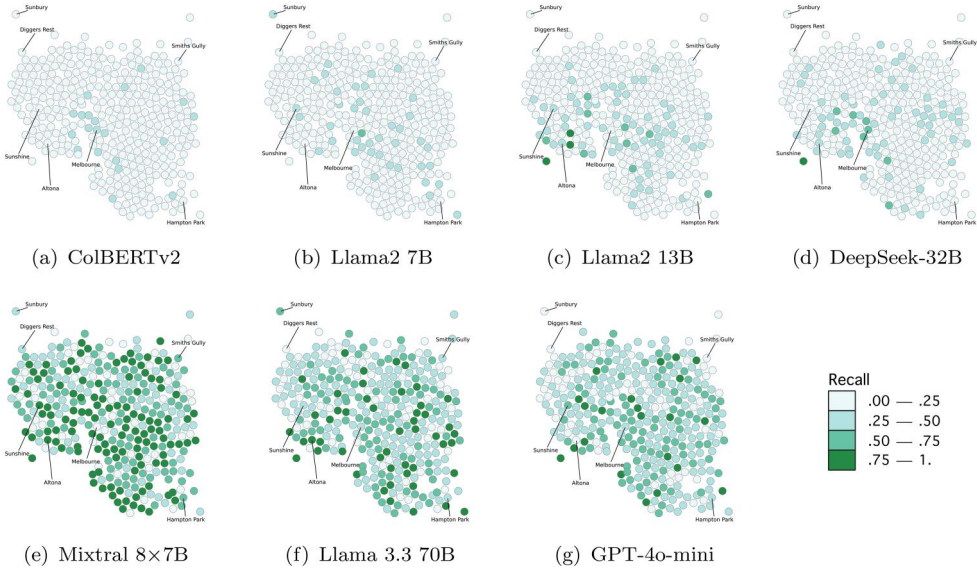


Figure 8. The suburb-level breakdown of *recall* for all compared language models.

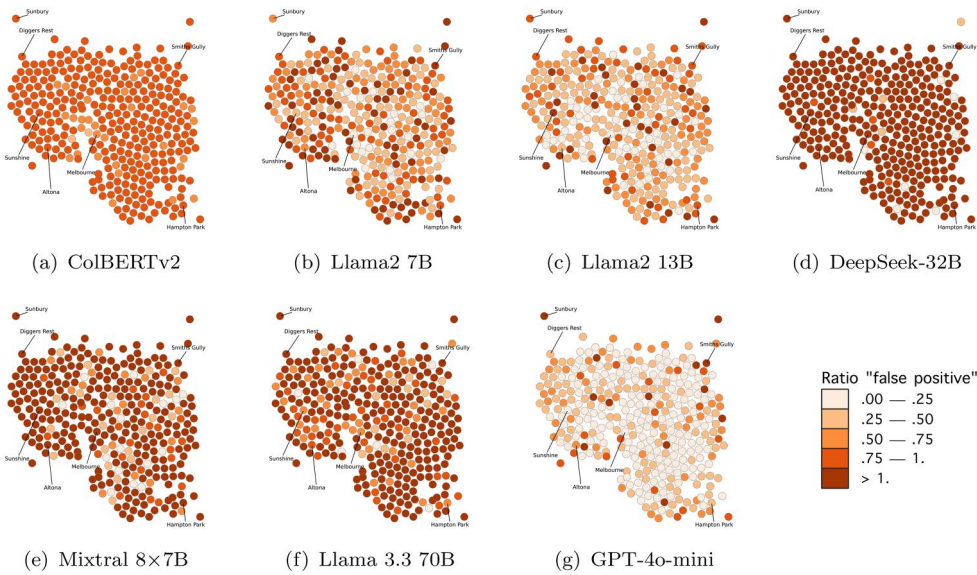


Figure 9. The suburb-level breakdown of the combined $FP_{-00v}/|G_S|$ and $FP_{00v}/|G_S|$ for all compared language models.

language models struggled to capture adequately the symmetry of extracted adjacency relations.

Figure 10 summarizes the suburb-level breakdown of symmetry. Note that in our results, symmetry was evaluated for *any* symmetric edge in G_{LM} , including false positive and false negative edges: evaluating symmetry on any edge (correct or not) measured how the models considered symmetry as a reciprocal relation (despite errors). The distribution of centrality values tended to be highly uneven across suburbs. Even

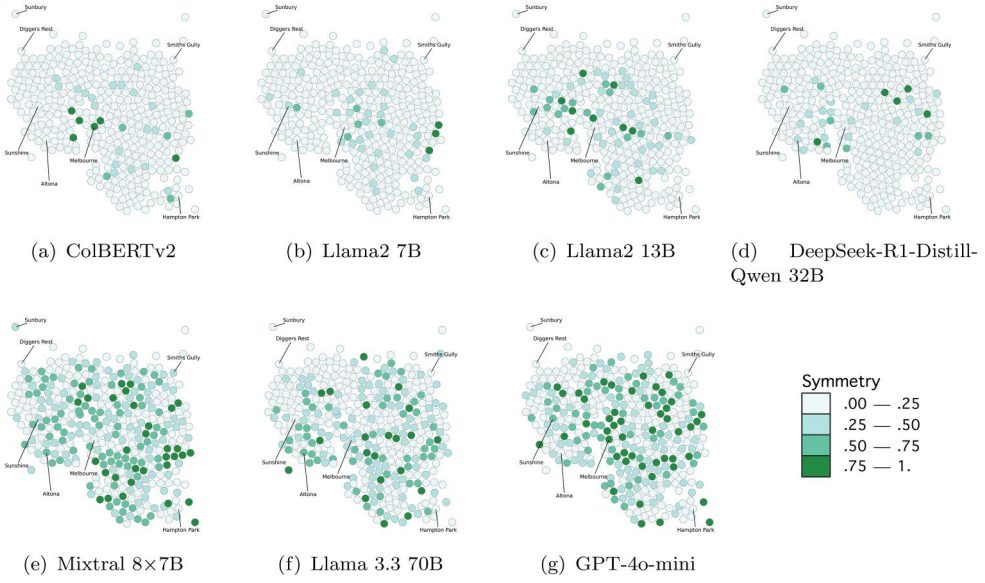


Figure 10. The suburb-level breakdown of *symmetry* for all compared language models.

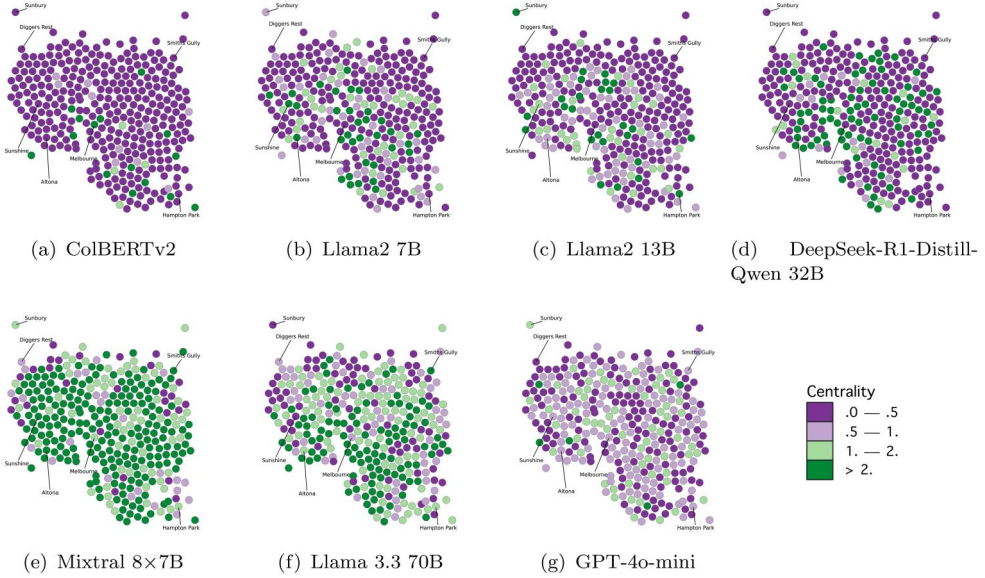


Figure 11. The suburb-level breakdown of *indegree centrality* for all compared language models. A ratio above 1 indicates a high-centrality node (“hotspot node”) receiving high number of incoming edges.

for the best performing model (GPT-4o-mini), only around 23% of the edges were symmetric.

A further spatial bias in language models was suggested by an examination of Figure 10, where symmetry was preserved only for certain nodes in the graph. To study this phenomenon, Figure 11 reports the indegree centrality per suburb. The results suggested a tendency for language models to overconnect nodes to a subset

of high-centrality suburbs. In particular, the suburbs around the historical settlement of Greater Melbourne (around the suburb “Melbourne City”) had a high centrality in most language models, ranging from smaller models such as ColBERTv2 to most recent LLM GPT-4o-mini.

5.6. Discussion

Language models involved in modular architectures had integrated some gazetted knowledge. However, their performance presented discrepancies when compared with the geographic ground truth.

Fundamentally, language models operated as degraded representations of gazetteers. Probing LLMs with the relation of “adjacency” revealed limitations in making spatio-textual connections. Despite the scaling effect of increased parameters, LLMs were unreliable spatial databases and disregarded spatial structures. Language models frequently generated fabricated or non-gazetted toponyms, and failed to preserve known spatial relations between places, such as adjacency or symmetry. These results aligned with recent studies, such as Renshaw *et al.* (2025) on US county-level errors; and Ji *et al.* (2025) who reported a limited accuracy of 0.6 for topological spatial relation inference with GPT-4, as well as confusion regarding RCC-8 relations.

Popularity bias further distorted geographic representations, effectively compressing the geographic domain into a preferential set of locations. The spatial distribution of errors was nonrandom, exhibiting isolated hotspots. In particular, centralized connections to the same suburbs could be associated with a popularity bias (Klimashevskaja *et al.* 2024). The phenomenon was more pronounced in smaller language models that tended to reduce space to a limited number of suburbs. This characteristic is an asset in recommendation, when prioritizing popular locations.

Crucially, LLMs treated the logic expressed through word regularities as soft constraints rather than hard logic. We observed that larger models tended to overconnect suburbs, creating a “hallucination” where distinct locations appeared artificially close. Consequently, larger language models had a distorted definition of adjacency: the spatial relation “adjacent to” was more akin to the vague concept “near to.” Prompt engineering is certainly a way to refine retrieval by describing more soft constraints for the adjacency relation, or any other relation in future work. Nonetheless, the conversion of soft constraints into hard constraints is incompatible with any model relying on probabilistic word associations rather than spatial structures.

6. Conclusions

Foundational data, such as gazetteers, are considered authoritative due to their wide use in most GIS applications. However, authenticating data origin is often reduced to the determination of a custodian, and rarely to the etymology with a record in the literature. Records of toponym origins serve as compelling benchmark datasets to evaluate how spatial text mining techniques perform in identifying data origin via either *factoid retrieval* or *candidate recommendation*. We demonstrated not only that

retrieving data origin is a challenging task, but also that the involved language models still lacked reliability when performing spatial matches.

In recent years, the integration of nonspatial tools (e.g., LLMs) has extended the capabilities of GIS, acting as end-to-end models to bypass highly engineered pipelines. Despite improved generalization, this simplification incurs information loss due to the use of non-spatial models. Language models are not necessarily well-adapted to domains beyond pure natural language, such as the spatial domain. Indeed, there is a gap between how language models spatially describe the world in words and how the world is actually partitioned in spatial concepts.

We observed two main factors that were exclusive to geographic information retrieval based on language models. First, language models exhibited spatial biases. The degree of bias created a sort of spatial resolution that limited the retrieval of elements at lower scales. Second, language models treated any constraint as a soft constraint, with more emphasis on semantic similarities to the detriment of spatial coherence: spatial grounding was easily missed in systems that lacked explicit spatial structure—a problem likely exacerbated in more complex systems, such as autonomous LLM-based agents. Consequently, spatial knowledge discovery remains an open problem with current language models.

Determining data origin is a crucial step in database management and trustworthiness. In interdisciplinary applications such as fact-checking Kotonya and Toni (2020), unsolved challenges remain in bridging the gap between word similarities and spatial relevance.

Notes

1. United Nations, Decision 9/111 in August 2019.
2. <https://www.land.vic.gov.au/place-naming/services-and-resources/vicnames-register> (26/03/2024).
3. <https://discover.data.vic.gov.au/dataset/register-of-geographic-place-names-vicnames-current>.
4. foursquare/fsq-os-places; bounded by $\text{lat} \in [-39.2^\circ, -34.0^\circ]$, $\text{lon} \in [140.9^\circ, 150.0^\circ]$.

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Author contributions

CRedit: **Alexis Horde-Vo**: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing; **Matt Duckham**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Estrid He**: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Nayomi Geethanjali Ranamuka**: Conceptualization, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review &

editing; **Rafe Benli**: Conceptualization, Validation, Writing – original draft, Writing – review & editing.

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Data and codes availability statement

The data and codes that support the findings of this study are available at: <https://doi.org/10.25439/rmt.30073789>

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Appendix A. Experiment 1: complete prompt for decoder models

The complete prompt used for decoder models is:

[Example] List all the suburbs that are immediately adjacent to Monaco. Delimitate each suburb with `<suburb>` `</suburb>`. Be concise. Write “# END” when you finish your answer. `<suburb>` Cap-d’Ail`</suburb>` `<suburb>` Beausoleil`</suburb>` `<suburb>` Roquebrune-Cap-Martin`</suburb>` [Task] List all the suburbs that are immediately adjacent to {city}, Victoria, Australia. Delimitate each suburb with `<suburb>` `</suburb>`. Be concise. Write “# END” when you finish your answer.

Appendix B. Experiment 2: Prompts

Given the sensitivity of language models to prompt engineering, we empirically selected the following prompts:

- Pr_{sem} : “Who is {street_name} most likely named after, in Melbourne, Victoria, Australia? If it is not a person, find any other origin.”
- Pr_{geo} : “What event occurred in Melbourne, Victoria, Australia? {instructions}”

For triplet extraction in the rewriter module, we used:

- $Pr_{extract}$: “Task - Keywords for mentions, locations and dates are given: `< mentions >` {list} `</mentions >` (...) Extract the triplets `< triplet >` object, relation, location `</triplet >` or `< triplet >` location, relation, object `</triplet >`. {instructions}”

For final decision-making, the generator received the top- k_{ranker} documents $D_{i,j}$ sorted in descending order of relevance to optimize the context for next-token generation:

- $Pr_{generator}$: “Who is {street_name} most likely named after, in Melbourne, Victoria, Australia? If it is not a person, find any other origin. For the top-{ k_{ranker} } knowledge graphs, an extract from the RDF/XML file is provided. These are the extracts: {top-10_documents_ $D_{i,j}$ } {instructions}.”

Appendix C. Ranking metrics

Let X_i be the i -th query ($i \in \{1, \dots, N\}$) and $y_i \in \{0, 1\}^k$ be its boolean relevance vector for the top- k recommended items, where $y_{i,j} = 1$ if the item at rank j is relevant, and 0 otherwise. The evaluation metrics are summarized in Table 3.

Table C1. Summary of ranking metrics at cutoff k .

Metric	Formula	Description
HR@ k	$\frac{1}{N} \sum_{i=1}^N \mathbb{I}(\sum_{j=1}^k y_{i,j} > 0)$	Proportion of queries with ≥ 1 relevant item in top- k .
P@ k	$\frac{1}{N} \sum_{i=1}^N \left(\frac{1}{k} \sum_{j=1}^k y_{i,j} \right)$	Average proportion of relevant items within the top- k .
DCG@ k	$\sum_{j=1}^k \frac{y_{i,j}}{\log_2(j+1)}$	Penalizes relevant items appearing lower in the ranking.
nDCG@ k	$\frac{1}{N} \sum_{i=1}^N \frac{\text{DCG}@k(y_i)}{\text{idCG}@k(y_i)}$	Average DCG normalized by the ideal maximum score for each query.

Appendix D. supplementary experiment 1b: Place recommendation

To validate the generalizability of our observations in Section 4 beyond toponymic resolution, we evaluated a place recommendation task. We adapted the parallel ranking mechanism (Equation 2, simplified to $D_{i,j} = D_{i,j}^{\text{geo}}$) to prioritize points of interest (POIs) based on semantic queries, such as $Q_{\text{sem}} = \text{"Where can I have lunch in [city]?"}$.

Ground-truth relevance was strictly prioritized by categorical hierarchies within the target city: (1) *Restaurants* (C_1 , ideal rank 1), (2) *Bars* (C_2 , ideal rank 10), and (3) *Retail/Other* (C_3 , ideal rank 100).

D.1. Experimental setup

Dataset. We extracted 20,000 POIs across 2,857 cities around Greater Melbourne from the Foursquare dataset⁴. Constraining the scope to Greater Melbourne mitigated global toponymic ambiguity and popularity biases. To control the distribution, C_1 and C_2 targets were capped at 10 POIs per city.

Methodology. POIs were formatted as standardized documents: $D = \text{"[place] is a [commercial function] located in [city]."}$ We benchmarked five text encoders (all-MiniLM-L6-v2 Reimers and Gurevych (2019), ColBERTv2 Santhanam *et al.* (2022), BGE base Chen *et al.* (2024), EmbeddingGemma Vera *et al.* (2025), and Qwen3 Yang *et al.* (2025)) evaluating the top-10 results across two ranker architectures:

1. **C2a (single ranker):** computed semantic relevance only, $\text{Sim}(Q_{\text{sem}}, D)$.
2. **V1 (parallel ranker):** computed the element-wise product of semantic and geographic relevance, $\text{Sim}(Q_{\text{sem}}, D) \times \text{Sim}(Q_{\text{geo}}, D)$, where $Q_{\text{geo}} = \text{"It is a place in [city]"}$.

D.2. Results

Table 4 reports Mean Reciprocal Rank (MRR), nDCG, Precision (P) and Hit Ratio (HR) for the evaluation. We observed the following:

Table D1. Recommendation performances.

Model/Label	Model	MRR	nDCG	P	HR
all-MiniLM-L6-v2	C2a	.39	.27	.22	.54
	V1	.50 (+29%)	.39 (+43%)	.36 (+64%)	.66 (+21%)
ColbertV2	C2a	.20	.10	.06	.31
	V1	.29 (+45%)	.16 (+53%)	.11 (+72%)	.41 (+31%)
BGE	C2a	.55	.46	.41	.70
	V1	.63 (+15%)	.57 (+23%)	.54 (+32%)	.80 (+16%)
Qwen3	C2a	.48	.37	.32	.64
	V1	.52 (+8%)	.41 (+9%)	.37 (+16%)	.68 (+7%)
EmbeddingGemma	C2a	.47	.39	.31	.66
	V1	.58 (+22%)	.50 (+30%)	.45 (+46%)	.76 (+16%)

- **Superiority of parallel ranking:** the V1 architecture consistently and significantly outperformed the single C2a ranker across all text encoders and metrics.
- **Spatial anchoring deficit:** even though geographic matching was reduced to an exact string match in this simplified setup (the target city was explicit in both query and document), absolute performance remained low, particularly for sorting quality (nDCG).

These findings corroborated our primary evaluation in [Section 4](#): language models processed spatial conditions as soft constraints rather than strict filters, confirming that robust spatial anchoring remains a critical challenge for location-based recommendation.